

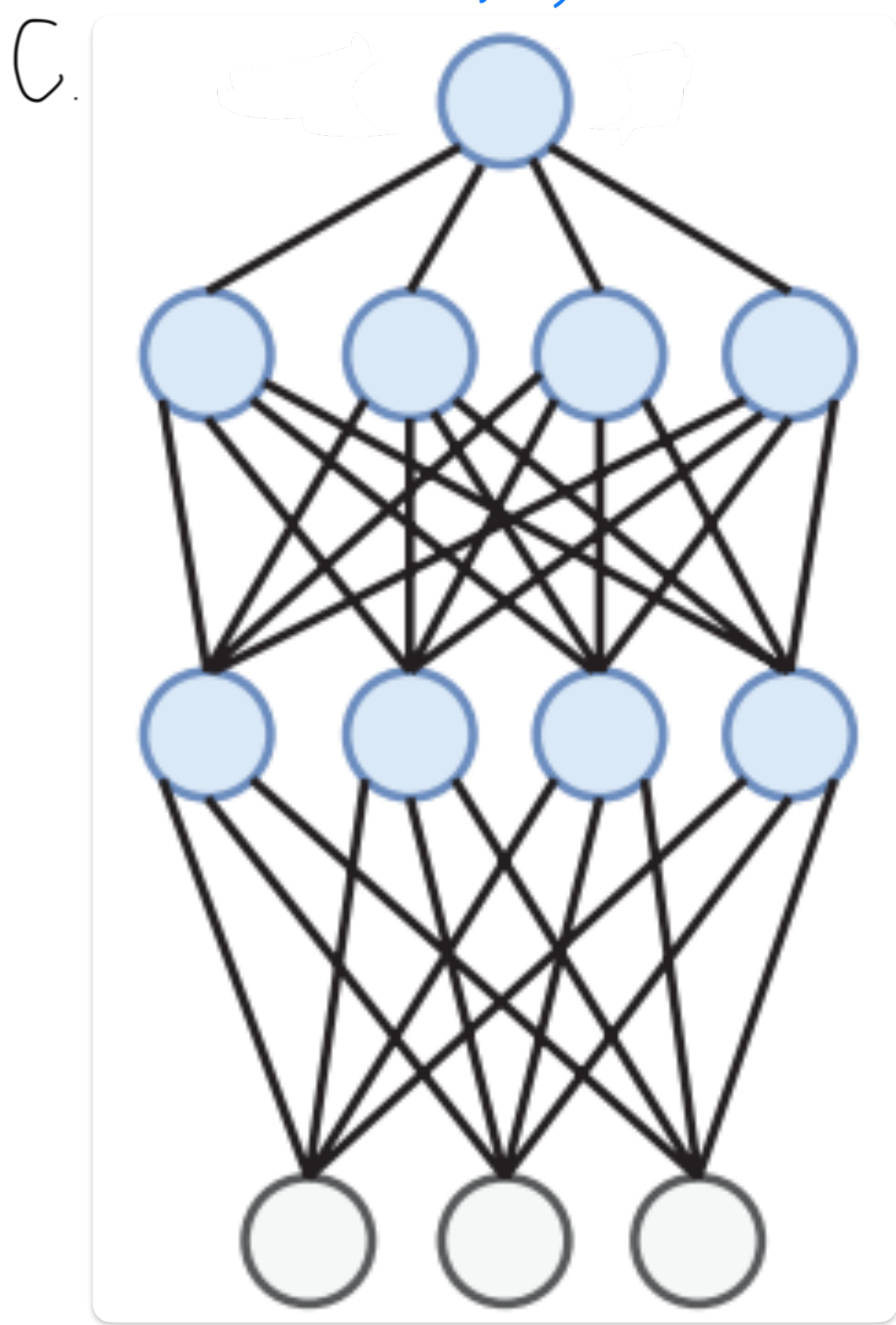
1. Network Architecture in Definition:

- A. network architecture show how multiple perceptrons are connected when we build complex neural network.
- B. 3 types of fundamental network architectures

- fully connected forward neural network
- convolutional neural network (CNN)
- recurrent neural network (RNN)

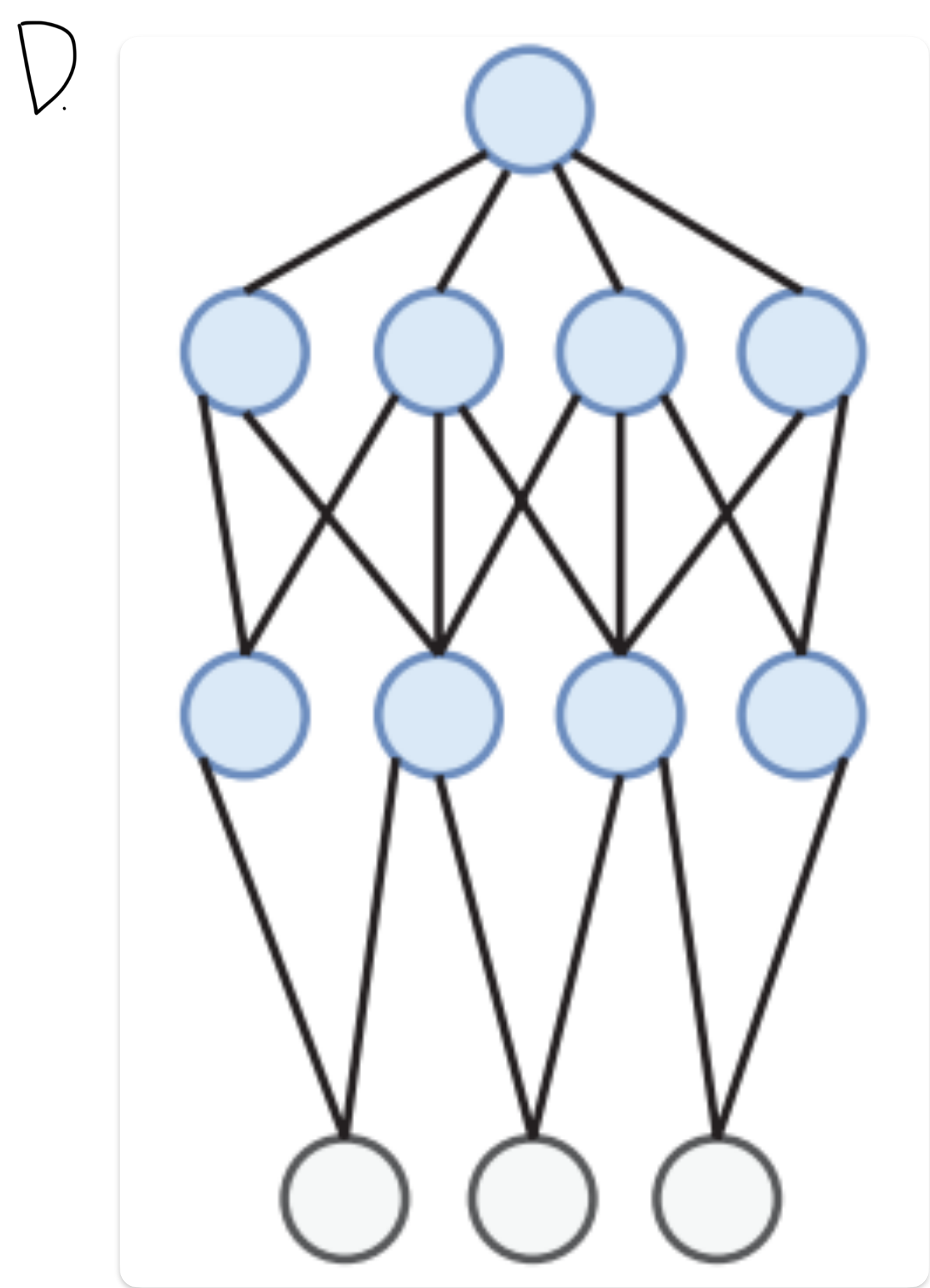
2. Fully Connected Forward Neural Network:

- A. in this neural network, there are no backward connections between perceptrons, which is the same characteristic as Directed Acyclic Graph (DAE) with no loop or cycle inside.
(no backward connection is about physical neural network structure, this neural network can be trained by Backpropagation)
- B. in this neural network, each of perceptrons in the first layer will connect to all perceptrons in the next layer without exception,
this is a property known as fully connected



3. Convolutional Neural Network (CNN):

- A. in this neural network, the individual perceptrons don't have their own unique weights,
they use the same weights as other perceptrons in the same layer;
this is a property known as weight sharing.
- B. in this neural network, it has considerably fewer connections between perceptrons than a fully connected neural network,
this is a property known as sparsely connected
- C. CNN have been shown to excel on image classification problems.



4. Recurrent Neural Network (RNN):

- A. in this neural network, there are not only forward connections but also backward connections between perceptrons, so this neural network contains loops or cycles inside and doesn't belong to Directed Acyclic Graph (DAG),
this design of backforward connection can feed outputs back into inputs, which gives the neural network the ability to retain previous state (memory),
and then applying this design on processing sequential context info.

